

Exam

Name _____

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

- 1) Which is NOT a pathogen? 1) _____
 - A) athlete's foot fungus
 - B) meningococcus bacterium
 - C) chicken pox virus
 - D) hookworm parasite
 - E) cancer

- 2) Autoimmune diseases are an example of a(n) _____ response by the immune system. 2) _____
 - A) lack of
 - B) incorrect
 - C) overactive
 - D) None of the answers are correct.

- 3) When the immune system creates a response that is out of proportion to the threat it has been exposed to, the response is called an 3) _____
 - A) allergy or hypersensitivity reaction.
 - B) autoimmune response.
 - C) immunodeficiency.

- 4) Antibiotics are drugs that destroy or inhibit the growth of 4) _____
 - A) bacteria.
 - B) viruses and bacteria.
 - C) viruses, bacteria, and multicellular pathogens such as hookworms.
 - D) multicellular pathogens such as hookworms.
 - E) viruses.

- 5) Viruses 5) _____
 - A) easily survive in the environment outside the body.
 - B) are parasitic.
 - C) must reproduce inside host cells.
 - D) must reproduce inside host cells, are parasitic, and easily survive in the environment outside the body.
 - E) must reproduce inside host cells and are parasitic.

- 6) Which is NOT matched correctly? 6) _____
 - A) active immunity – an immune response started and developed by the cells
 - B) adaptive immunity – responses directed toward specific invaders
 - C) innate immunity – nonspecific responses
 - D) active adaptive immunity – immunity passed through the placenta or milk to offspring
 - E) Each of these is correctly described.

- 7) Lymphoid tissues include 7) _____
 - A) lymph nodes.
 - B) spleen.
 - C) bone marrow.
 - D) thymus.
 - E) All of the answers are correct.



- 8) The largest lymphoid organ(s) is/are the _____
 A) GALT.
 B) thymus.
 C) lymph nodes.
 D) tonsils.
 E) spleen.
- 9) Diffuse lymphoid tissue includes _____
 A) GALT (gut-associated lymphoid tissue).
 B) tonsils, GALT (gut-associated lymphoid tissue), and the spleen.
 C) tonsils.
 D) the spleen.
 E) tonsils and GALT (gut-associated lymphoid tissue).
- 10) Swelling and soreness of the lymph nodes indicate that _____
 A) immune cells are fighting infection.
 B) dangerous cancerous changes are occurring in the immune system.
 C) the immune system is incapable of working.
 D) too much or improper exercise has been undertaken.
 E) None of the answers are correct.
- 11) _____ are clusters of lymphatic nodules located beneath the epithelial lining of the small intestine. _____
 A) Lymph nodes
 B) Tonsils
 C) Peyer's patches
 D) Adenoids
 E) Immune complexes
- 12) Cells of the mononuclear phagocyte system include _____
 A) macrophages and monocytes.
 B) macrophages, monocytes, neutrophils and eosinophils.
 C) macrophages, monocytes, neutrophils, eosinophils, lymphocytes and plasma cells.
 D) lymphocytes and plasma cells.
 E) neutrophils and eosinophils.
- 13) Which cell is NOT correctly matched with the characteristic? _____
 A) eosinophils – numbers increase with allergies and parasites
 B) eosinophils – cytotoxic cells may live only 6–12 hours
 C) neutrophils – most difficult to identify
 D) basophils – have granules
 E) neutrophils – most abundant white cell
- 14) _____ are the precursor cells of macrophages. _____
 A) Eosinophils
 B) Basophils
 C) Monocytes
 D) Lymphocytes
 E) Neutrophils



- 15) Which are NOT antigen-presenting cells (APCs)? 15) _____
 A) dendritic cells
 B) neutrophils
 C) monocytes
 D) macrophages
 E) lymphocytes
- 16) Leukocytes recognize molecules that are unique to pathogens; thus, the molecules are 16) _____
 A) antigens.
 B) pathogen-associated molecular patterns.
 C) pattern recognition receptors.
 D) toll-like receptors.
 E) phagosomes.
- 17) Chemical signals that attract immune cells are known as 17) _____
 A) opsonins.
 B) chemotaxins.
 C) lysozymes.
 D) antigens.
 E) interferons.
- 18) Which immune cells are also known as Langerhans cells? 18) _____
 A) mast cells
 B) eosinophils
 C) basophils
 D) dendritic cells
 E) lymphocytes
- 19) Lymphocytes and plasma cells 19) _____
 A) mediate the adaptive immune response.
 B) all function the same way.
 C) are found mostly in lymphoid tissue.
 D) mediate the adaptive immune response and are found mainly in lymphoid tissue.
 E) mediate the adaptive immune response, are found mainly in lymphoid tissue, and all function the same way.
- 20) C-reactive protein is a type of 20) _____
 A) pyrogen.
 B) cytokine.
 C) acute-phase protein.
 D) granzyme.
 E) chemotaxin.
- 21) Which is NOT a characteristic of adaptive immunity? 21) _____
 A) acquired immunity
 B) cell-mediated immunity
 C) humoral immunity
 D) nonspecific immunity



- 22) Cytokines function by 22) _____
 A) attracting immune cells and causing fever and poisoning and killing invading cells.
 B) attracting immune cells and causing fever, poisoning and killing invading cells, and digesting invading cells.
 C) poisoning and killing invading cells.
 D) attracting immune cells and causing fever.
 E) digesting invading cells.
- 23) Chemotaxins function in 23) _____
 A) digesting invading cells, attracting additional immune cells, and poisoning and killing invading cells.
 B) digesting invading cells.
 C) attracting additional immune cells.
 D) poisoning and killing invading cells.
 E) digesting invading cells and attracting additional immune cells.
- 24) A substance that induces fever is known as a(n) 24) _____
 A) acute phase protein.
 B) chemotaxin.
 C) cytokine.
 D) opsonin.
 E) pyrogen.
- 25) Opsonins function by 25) _____
 A) marking or tagging pathogens so phagocytes can find and ingest them.
 B) poisoning and killing invading cells and marking or tagging pathogens so phagocytes can find and ingest them.
 C) poisoning and killing invading cells, marking or tagging pathogens so phagocytes can find and ingest them, and digesting invading cells directly.
 D) digesting invading cells directly.
 E) poisoning and killing invading cells.
- 26) Acute phase proteins are produced by the 26) _____
 A) macrophages.
 B) liver.
 C) thymus.
 D) lymphocytes.
 E) lymph nodes.
- 27) The cells responsible for the production of circulating antibodies are 27) _____
 A) helper T cells.
 B) cytotoxic T cells.
 C) NK cells.
 D) suppressor T cells.
 E) plasma cells.
- 28) Passive immunity is a type of _____ immunity. 28) _____
 A) innate B) adaptive



- 29) The cells responsible for cell-mediated immunity are the _____
 A) T lymphocytes. B) B lymphocytes.
 C) suppressor T lymphocytes. D) plasma cells.
- 30) Stem cells that will form B lymphocytes are found in the _____
 A) kidneys.
 B) bone marrow.
 C) spleen.
 D) thymus.
 E) liver.
- 31) The initial cellular defense against pathogens are the _____
 A) plasma cells.
 B) phagocytes.
 C) B lymphocytes.
 D) NK cells.
 E) T lymphocytes.
- 32) Which acts as first line of defense against pathogens? _____
 A) inflammation B) skin C) macrophages D) lymphocytes
- 33) An inflammatory response is triggered when _____
 A) T lymphocytes release interferon.
 B) red blood cells release pyrogens.
 C) blood flow to an area increases.
 D) neutrophils phagocytize bacteria.
 E) mast cells release granules containing histamine.
- 34) Lymphocytes that attack body cells infected with viruses are _____
 A) B lymphocytes.
 B) helper T cells.
 C) cytotoxic T cells.
 D) suppressor T cells.
 E) plasma cells.
- 35) Cells that help to influence other immune cells are _____
 A) NK cells.
 B) helper T cells.
 C) B lymphocytes.
 D) cytotoxic T cells.
 E) plasma cells.
- 36) The cells that are actively involved in immune surveillance are the _____
 A) plasma cells.
 B) helper T cells.
 C) suppressor T cells.
 D) NK cells.
 E) B lymphocytes.



- 37) Stem cells that will form T lymphocytes mature in the _____
 A) thymus.
 B) liver.
 C) spleen.
 D) kidneys.
 E) bone marrow.
- 38) _____ is the immunoglobulin class that make up 75% of antibodies in adult blood. _____
 A) IgA B) IgD C) IgE D) IgG E) IgM
- 39) Immunoglobulins that cross the placenta are _____
 A) IgA. B) IgD. C) IgE. D) IgG. E) IgM.
- 40) Immunoglobulins that attach to mast cells and are involved in allergic reactions are _____
 A) IgA. B) IgD. C) IgE. D) IgG. E) IgM.
- 41) Immunoglobulins that are found on the surface of B lymphocytes and may play a role in regulation of the humoral immune response are _____
 A) IgA. B) IgD. C) IgE. D) IgG. E) IgM.
- 42) Immunoglobulins in early responses to pathogens are _____
 A) IgA. B) IgD. C) IgE. D) IgG. E) IgM.
- 43) Immunoglobulins in external secretions are _____
 A) IgA. B) IgD. C) IgE. D) IgG. E) IgM.
- 44) The specificity of an antibody is determined by the _____
 A) antibody class.
 B) fixed segment.
 C) variable region.
 D) size of the antibody.
 E) antigenic determinants.
- 45) For a lymphocyte to respond to an antigen, the antigen must _____
 A) be phagocytized by the lymphocyte.
 B) bind to the DNA of the lymphocyte.
 C) enter the cytoplasm of the lymphocyte.
 D) depolarize the lymphocyte.
 E) bind to specific receptors on the lymphocyte membrane.
- 46) When an antigen fragment is bound to a MHC class I molecule to form a MHC-I-antigen complex, it can stimulate a _____
 A) NK cell.
 B) cytotoxic T cell.
 C) B cell.
 D) plasma cell.
 E) helper T cell.



- 47) MHC class II molecules are found 47) _____
 A) only on red blood cells.
 B) only on granulocytes and macrophages.
 C) only on dendritic cells and macrophages.
 D) on all cells with a nucleus.
 E) only on liver cells and macrophages in the spleen.
- 48) A person's blood type is determined by the 48) _____
 A) number of specific molecules on the cell membrane.
 B) chemical character of the hemoglobin.
 C) shape of the red blood cells.
 D) size of the red blood cells.
 E) presence or absence of specific glycoprotein molecules on their cell membrane.
- 49) A person with type A blood has 49) _____
 A) anti-B antibodies in his plasma.
 B) anti-A antibodies on his red blood cells.
 C) the ability to receive AB blood cells.
 D) anti-B antibodies on his red blood cells.
 E) anti-A antibodies in his plasma.
- 50) Overactive immune responses to antigens are 50) _____
 A) immunodeficiency diseases.
 B) allergies and hypersensitivity reactions.
 C) the result of stress.
 D) autoimmune diseases.
 E) common in the elderly.

MATCHING. Choose the item in column 2 that best matches each item in column 1.

Match the cell type with its description.

- | | | |
|---|--------------------------------------|-----------|
| 51) These are the most abundant leukocyte; they are formed in the bone marrow. | A) neutrophils

B) eosinophils | 51) _____ |
| 52) Cytotoxic, these cells are associated with allergic reactions and parasitic infections. | | 52) _____ |



- | | | |
|---|----------------|-----------|
| 53) Precursors of macrophages, these cells are relatively rare in blood. | A) basophils | 53) _____ |
| | B) lymphocytes | |
| 54) These cells are related to mast cells, and both release chemical mediators, which aid in the inflammatory and immune responses. | C) monocytes | 54) _____ |
| 55) Estimated to reach populations of one trillion in normal adults, most of these cells are found in lymphoid tissues. These cells look similar to one another under the microscope, but function differently. | | 55) _____ |

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

- | | |
|--|-----------|
| 56) A patient has increased eosinophils and monocytes but normal levels of neutrophils, basophils, and lymphocytes. Which is the most likely cause of these test results? | 56) _____ |
| A) a chronic parasitic infection
B) a chronic bacterial infection
C) a viral infection
D) a severe allergic reaction
E) an acute bacterial infection | |
| 57) A patient has increased neutrophils and macrophages with normal levels of basophils, eosinophils, and lymphocytes. Which is the most likely cause of these test results? | 57) _____ |
| A) an acute viral infection
B) an allergic reaction
C) an autoimmune reaction
D) a fungal infection
E) a chronic bacterial infection | |
| 58) A patient with hay fever symptoms for a couple weeks comes to your office to be treated for his hay fever. If a white blood cell differential count was performed, which leukocytes would likely be increased? | 58) _____ |
| A) neutrophils, monocytes, and lymphocytes
B) monocytes, eosinophils, and basophils
C) lymphocytes, basophils, and neutrophils
D) monocytes, lymphocytes, and eosinophils | |

MATCHING. Choose the item in column 2 that best matches each item in column 1.

Match the type of immunity to its description.

- | | | |
|---------------------------------------|-----------|-----------|
| 59) present at birth | A) innate | 59) _____ |
| 60) Secreted lysozyme kills bacteria. | | 60) _____ |



- | | | |
|--|-------------|-----------|
| 61) NK cells cause apoptosis of viral-infected cells. | A) adaptive | 61) _____ |
| | B) innate | |
| 62) Antibodies in breast milk protect a newborn baby from pathogens. | | 62) _____ |
| 63) Infected cells are killed by perforin and granzymes. | | 63) _____ |

SHORT ANSWER. Write the word or phrase that best completes each statement or answers the question.

- | | |
|---|-----------|
| 64) _____ macrophages are phagocytic cells that are permanent residents of specific tissues and organs. | 64) _____ |
| 65) _____ are fixed macrophages found in the brain. | 65) _____ |
| 66) _____ are fixed macrophages found in the liver. | 66) _____ |
| 67) _____ are dendritic cells found in the skin. | 67) _____ |
| 68) _____ are free phagocytes found in the lungs. | 68) _____ |
| 69) The ability of certain cells to mobilize in response to changes in their chemical environment is called _____. | 69) _____ |
| 70) Substances that trigger an immune response are known as _____. | 70) _____ |
| 71) _____ are antibodies. | 71) _____ |
| 72) The process by which the surface of a microorganism is covered, making it more likely to be phagocytized occurs when _____ act as opsonins. | 72) _____ |
| 73) _____ cells enable the immune system to respond more quickly if the same antigen is encountered a second time. | 73) _____ |
| 74) _____ exists when the immune system does NOT respond to a particular antigen on cells of the body. | 74) _____ |
| 75) _____ are mediators released from white blood cells that act on other white blood cells and mediates innate defenses. | 75) _____ |
| 76) _____ are cytokines that inhibit the replication of viruses. | 76) _____ |
| 77) _____ are groups of lymphocytes bind to a particular ligand. | 77) _____ |



MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

- 78) A sample of tissue from an injury shows a large number of basophils. This would indicate that the tissue was 78) _____
 A) being rejected.
 B) abscessed.
 C) inflamed.
 D) infected by multicellular parasites.
 E) infected by viruses.
- 79) Meghan thinks she has an abscessed tooth. If she does, which type of white blood cell would you expect to see in increased numbers in a differential count? 79) _____
 A) eosinophils
 B) lymphocytes
 C) basophils
 D) monocytes
 E) neutrophils
- 80) Ricardo wants to determine his blood type, so he takes a few drops of blood from his finger and mixes it with various antibodies. His blood cells agglutinate when mixed with the anti-A antibody but not with the anti-B antibody. Therefore, which is true? 80) _____
 A) His plasma would cross-react with type O red blood cells.
 B) His plasma contains B antibodies.
 C) He could receive type B blood in a transfusion.
 D) He is Rh positive.
 E) He could donate blood to an individual with type B blood.
- 81) In an experimental situation focused on the development of a new vaccine, mice were injected with a viral antigen. Their plasma was removed in order to detect the presence of antibodies. This is an example of 81) _____
 A) innate immunity.
 B) passive immunity.
 C) active immunity.
 D) natural immunity.
 E) autoimmunity.
- 82) In an experimental situation, a virus is injected into a rabbit and the rabbit is allowed to make antibodies against the viral antigen. These antibodies are then removed from the rabbit plasma and injected into a human to help fight the same viral disease. This is an example of 82) _____
 A) passive immunity.
 B) active immunity.
 C) autoimmunity.
 D) innate immunity.
 E) natural immunity.
- 83) A decrease in which lymphocytes would decrease all parts of an immune response? 83) _____
 A) helper T cells
 B) B lymphocytes
 C) regulatory T cells
 D) plasma cells
 E) cytotoxic T cells



- 84) The human immunodeficiency virus (HIV) that causes the disease known as AIDS selectively infects 84) _____
- A) suppressor T cells.
 - B) cytotoxic T cells.
 - C) plasma cells.
 - D) helper T cells.
 - E) B cells.
- 85) Milly has just received a kidney transplant and is taking cyclosporin A, a drug that inhibits cytotoxic T cells. What does this medication do? 85) _____
- A) prevents rejection
 - B) prevents inflammation from destroying the transplanted kidney
 - C) decreases hematopoiesis
 - D) increases the number of antibodies in the blood
 - E) None of the answers are correct.
- 86) Athena has a bad sore throat and the lymph nodes in her neck are swollen. This indicates 86) _____
- A) the focus of the infection is the lymph nodes.
 - B) the affected lymph nodes contain an increased number of lymphocytes.
 - C) the lymph gland has increased its secretion of thymosin.
 - D) lymph is not flowing through these lymph nodes.
- 87) Anil accidentally swallows a substance that activates the complement cascade in the absence of bound antibodies. This substance would 87) _____
- A) produce a fever.
 - B) increase circulation of lymph.
 - C) cause extensive cellular damage.
 - D) prevent inflammation.
 - E) activate blood clotting.
- 88) In a routine examination, some blood is taken and analyzed. The results show a high IgM level for the mumps antigen. This would indicate the 88) _____
- A) person is immune to mumps.
 - B) person is just coming down with mumps.
 - C) person is allergic to mumps.
 - D) person has just recovered from mumps.
 - E) None of the answers are correct.
- 89) A fluid sample contains a large amount of IgA type antibody. This fluid is probably 89) _____
- A) plasma.
 - B) intracellular.
 - C) lymph.
 - D) tears.
 - E) blood.



MATCHING. Choose the item in column 2 that best matches each item in column 1.

Match the immunoglobulin to its description.

90) crosses the placenta	A) IgA	90) _____
91) present in breast milk	B) IgG	91) _____
92) receptors on B lymphocytes	C) IgM	92) _____
93) binds to mast cells, triggering degranulation	D) IgD	93) _____
94) reacts to blood group antigens	E) IgE	94) _____

ESSAY. Write your answer in the space provided or on a separate sheet of paper.

- 95) Draw and label a typical antibody molecule. Be sure to include the light chain, heavy chain, Fab, Fc, and the hinge region. Identify the antigen-binding site and disulfide bonds.
- 96) Describe differences between bacteria and viruses.
- 97) Compare and contrast the effectiveness of the immune response to bacterial and viral infections.
- 98) Explain the role of exposure to antigens in the development of the immune system in the fetus and in the primary and secondary immune responses in children and adults. How does this compare to the role of innate immunity?
- 99) Describe innate and adaptive immunity.
- 100) Describe the role of complement in immunity.
- 101) Explain clonal expansion and the roles and interactions of effector cells, memory cells, naive lymphocytes, and plasma cells.
- 102) Explain the roles and interactions of each of the following in immunity: B lymphocytes, antibodies, immunoglobulins, T lymphocytes, MHC, HLA.
- 103) Explain each of these terms: *antibiotic, antibody, antigen, antihistamine, antiviral drug*.
- 104) Draw a concept map illustrating the immune response to bacterial infection.
- 105) Draw a concept map illustrating the immune response to a viral infection.
- 106) Describe the various functions of antibodies in an immune response.
- 107) Describe how a virus is inhibited in an immune response.
- 108) Describe the functions of cytotoxic T cells and NK cells.



- 109) The retrovirus HIV frequently mutates, making it difficult for scientists to develop a single effective vaccine. When flu vaccines are administered, patients are told that they are only protected from one strain or a few strains of the flu and are still as susceptible to other strains as unvaccinated people. Explain why.
- 110) Three different patients go in to see their gynecologist for some persistent, bothersome symptoms: Betelhem, Quinnie, and Armelle. Betelhem's doctor has detected benign uterine fibroid tumors. Quinnie's doctor has detected early-stage malignant vaginal tumors. Armelle's doctor has detected late-stage malignant ovarian tumors. Which part of the immune system would respond to these conditions? What would analysis of the nearby lymph nodes of these women show? What is the prognosis for each woman? Explain.
- 111) Jens has just been exposed to the measles virus, and he cannot remember if he has had measles before. He asks you if there is any way he can tell if he has been previously exposed or if he is going to get sick. What would you tell him? Why won't he get sick from a second infection?
- 112) Compare and contrast inflammation and infection.
- 113) Zoonotic diseases are transmissible from animals to humans. These include ringworm, a fungus which infects pets and people; plague, a bacterial disease resulting from a flea bite to mammals including humans; and rabies, a viral disease of the central nervous system that affects all mammals and is almost always fatal. On the other hand, the feline immunodeficiency virus has no effect on humans, and human immunodeficiency virus has no effect on cats, though each may be deadly to its normal host. Describe what may prevent cross-species infection in some cases and allow it in others.
- 114) Barun is 6 months old and requires a blood transfusion; his blood type is B. What type of blood may he receive? Is his age a factor in your answer? What would happen if he received a "wrong" blood type? If Barun were younger, is it possible that there is no "wrong" blood type? Why may it surprise a beginning physiology student, familiar with the basics of immunology, that a person could mount an immediate immune response to transfused blood? Explain all of your answers.
- 115) The HIV blood test detects anti-HIV antibodies. Why do viruses stimulate antibody production? How effective are antibodies against viruses, as compared to bacteria? How are viral infections normally eliminated in the body? Why does a person's immune response fail to eliminate life-long viral infections such as HIV and herpes? Explain all of your answers.
- 116) What do virus-infected cells and cancer cells have in common, related to the type of immune response and its long-term effectiveness?
- 117) The rejection of a transplanted organ by the immune system of the recipient is a result of a cell-mediated immune response, rather than antibody-mediated response. What does this tell you about the transplanted cells? From the perspective of the T cell, are the cells of the transplanted organ significantly different from virus-infected or cancerous cells?
- 118) Sketch the graphs of plasma antibody concentration in a primary immune response and a secondary immune response. Explain the graphs.



- 119) Shannon just finished finals and is heading home for winter break, excited for all the holiday activities her family has planned. It was a really stressful week and she just wants to get home to her family and friends. The day after finals Shannon wakes up feeling very sick with a cold, sore throat and muscle aches, probably the flu. Her father says the stress from finals probably caught up with her and she just needs to rest. What do you think? Is Shannon's father right? Could stress have caused her to get sick? If not, why not? If so, what mechanisms were at play?

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

- 120) Which represents the body's first line of defense to a pathogen? 120) _____
 A) B cells B) leukocytes C) stomach acid D) neutrophils

ESSAY. Write your answer in the space provided or on a separate sheet of paper.

- 121) Place the four basic steps of the immune response in the correct order.

MULTIPLE CHOICE. Choose the one alternative that best completes the statement or answers the question.

- 122) Which correctly explains the relationship between the nervous system, endocrine system, and immune system? 122) _____
 A) Cytokines from the immune system can affect neuroendocrine function.
 B) Only the nervous system and immune system share common receptors.
 C) Only the nervous system and endocrine system share common signal molecules.
 D) Hormones and neuropeptides do not alter the function of immune system cells.

